

Sangwon Lee

Robotics & Vision Software Engineer

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PROFESSIONAL SUMMARY

Robotics and vision software engineer with over 6 years of combined graduate research and industrial deployment experience, currently developing humanoid-robot vision and policy software at Samsung Electronics' Robotics AI Team (RX) after delivering production robot-vision and AI inspection systems for the Visual Display business. Hands-on across the full robotics-AI stack: VLA (Vision-Language-Action) model training on multi-GPU clusters, RL (reinforcement learning)-based policy post-training, real-time policy execution and speed acceleration, 2D/3D vision, instance segmentation and keypoint detection, and classical computer vision. Comfortable owning algorithm design through to deployment to live production lines across overseas plants.

PROFESSIONAL EXPERIENCE

Vision & Policy Software Engineer, Humanoid Robotics Jan 2026 – Present

Samsung Electronics — Robotics AI Team (RX), Seoul

Vision and policy-learning software for humanoid manipulation.

- **VLA (Vision-Language-Action) model development:** Designed a multi-modal fusion architecture that augments an open-source VLA backbone with an additional vision encoder through a learned projector, and implemented staged training for it on multi-node GPU clusters. Built a diagnostic suite that compares fused and baseline checkpoints across feature, action, and attention dimensions.
- **RL-based policy post-training:** Built an offline advantage-conditioned policy learning pipeline for a bimanual manipulation task, covering value estimation, advantage labeling, and conditioned fine-tuning, with evaluation gates that quantify how much signal the learned critic actually provides. Also investigated diffusion-steering RL and surveyed recent RL-for-VLA methods.
- **VLA execution-speed acceleration:** Surveyed and classified acceleration methods across inference latency, demonstration speedup, policy-level RL, and control to select an implementation target. Implemented a real-time speed-adaptation and tracking module built on optimization-based post-processing of policy output, validated in simulation and deployed to a physical robot.
- **Demonstration speedup & data quality:** Trained and evaluated speed-augmented policy checkpoints from time-warped demonstrations. Diagnosed timestamp desynchronization between sensor streams in the training data via cross-correlation, measured its effect on predicted actions, and designed a correction stage for the data conversion pipeline.
- **Detection with custom keypoint head:** Extended a single-stage detector to output object detections and 2D keypoints from one forward pass, designing the keypoint head and the multi-task training pipeline from data collection through evaluation.
- **Classical computer vision:** Designed robust geometric algorithms for structured environments — homography estimation under partial visibility, self-calibration, multi-stage RANSAC variants, and efficient line detection — as interpretable, fail-safe alternatives to learned models where reliability matters most.

Robot & Quality Control Vision System Engineer Sep 2022 – Jan 2026

Samsung Electronics — Visual Display, Manufacturing Process Engineering Group, Suwon

Robot vision and AI inspection systems deployed across global TV production lines.

- **Robot bin-picking vision:** Built the perception pipeline for a bin-picking cell that replaced existing parts feeders and removed a manual pre-alignment step, combining structured-light 3D capture, instance segmentation, pick-candidate selection, and plane fitting over the selected region to produce the grasp pose sent to the robot.
- **Palletizing robot vision:** Automated loading and unloading of product boxes, reducing logistics labor. Detected boxes from ToF depth input, recovered 3D target positions for the robot, and generated optimal stacking patterns for loading.

- **Robot 2D position-correction vision program:** Developed the vision program for industrial robots performing fastening, placement, and pressing tasks. Built an in-house template-matching algorithm with a hierarchical search structure that raises angular resolution only for high-similarity windows, and an in-house camera-to-robot 2D calibration. Parallelized communication and computation so that a single PC drives several robots and cameras concurrently, and designed the PC-to-controller communication protocol.
- **Stereo vision:** Estimated real-world 3D positions of fiducial markers from a two-camera rig using Direct Linear Transformation, and built the robot-controller communication module required for calibration.
- **Visual inspection system:** Built an inspection program covering surface defects by classification and assembly gaps by segmentation. Extracted regions of interest automatically from a multi-camera flow-shot setup and parallelized per-cycle deep-learning inference to fit the line cycle-time budget, with results reintegrated to their original positions on the operator UI. MLOps transition in progress.

EDUCATION

M.S. in Mechanical Engineering • Yonsei University, Seoul, South Korea Mar 2020 – Feb 2022

Precision Control Systems Laboratory (Mservo Lab), Advisor: Prof. Hyunseok Yang.

Thesis: Slip detection algorithm for robotic grippers using only normal-force sensor-array data — published in *Sensors and Materials* (SCI).

Lab project: Bilateral force-feedback teleoperation system pairing a hand-shaped exoskeleton (with Vive Tracker) and a remote robot arm — built SPI-based ADC modules for magnetic encoders and force sensors, Robotis Dynamixel actuator drivers, and a Gaussian-based shear-force estimation algorithm validated against ATI mini45 F/T ground truth.

B.S. in Mechanical Engineering • Yonsei University, Seoul, South Korea Mar 2014 – Feb 2020

PUBLICATIONS

- H. Bamshad, S. Lee, K. Son, H. Jeong, G. Kwon, H. Yang. "Multilayer-perceptron-based Slip Detection Algorithm Using Normal Force Sensor Arrays." *Sensors and Materials*, Vol. 35, No. 2, pp. 365–376 (2023).
- S. Lee, W. Jeon, K. Son, H. Yang. "Object Slip Detection Algorithm with the Sole Use of Normal Reaction Force Data." Korean Society of Mechanical Engineers (KSME) Honam Branch Conference, Poster (2021).

TECHNICAL SKILLS

Languages: Python, C++, C# (.NET), KUKA Robot Language

Algorithms & DSA: C++ data structures and algorithms; graph algorithms; time-constrained problem solving

AI / Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, LeRobot; YOLO11, Mask2Former, DETR, SigLIP, MLP / FCN / ResNet / Time-CNN / LSTM / LSTM-AutoEncoder, GAN, BERT; multi-node GPU cluster training (DDP / torchrun)

Policy Learning & Real-Time Execution: Real-Time Chunking, asynchronous inference, OSQP quadratic-programming solver, diffusion-steering RL (DSRL)

VLA & RL Literature: π_0 , $\pi_{0.5}$, $\pi_{0.6}^*$, $\pi_{0.7}$, OpenVLA, RT-2 / RT-X, RoboVLMs, EF-VLA, GR00T N1, Helix, Realtime-VLA, SAIL, DemoSpeedup, SpeedAug; DQN, GPS, DPG, DDPG, NPG, TRPO, PPO; David Silver RL lecture series

Computer Vision: OpenCV; hierarchical template matching, homography estimation, RANSAC, Direct Linear Transformation, camera-to-robot calibration; Zivid structured-light, Intel RealSense ToF, Basler stereo cameras, ArUco markers

Robotics: ROS, ROS2, forward / inverse kinematics, Gazebo simulation, path planning, KUKA robots

Control: LQR, LQG, MRAC, MPC, discrete PID (sine-sweep system identification), Kalman / low-pass / high-pass filters

Sensors / Actuators: Honeywell FSS force sensors, ATI mini45 F/T, magnetic encoders, custom piezo arrays, Robotis Dynamixel

Communication: I2C, SPI, TCP/IP, Modbus, CAN, PLC; multi-PC / multi-robot / multi-camera parallel I/O

Tools & Workflow: Linux, Docker; Git, Confluence, Jira; MATLAB / Simulink; SolidWorks / Inventor / Creo

Languages (Spoken): Korean (native); English (fluent — OPI: Superior)